```python
import seaborn as sns
import pandas as pd
import numpy as np
import tensorflow as tf
from sklearn.ensemble import StackingClassifier
from sklearn.metrics import roc_auc_score,RocCurveDisplay
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense, Dropout
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import RandomizedSearchCV
from sklearn.naive_bayes import MultinomialNB
from sklearn.preprocessing import StandardScaler
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import  OneHotEncoder
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay ,
classification_report,accuracy_score,precision_score,recall_score,f1_s
core
from tensorflow.keras.preprocessing.text import Tokenizer
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
import nltk
from nltk.corpus import stopwords
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

```
True
```

```python
df = pd.read_csv('fake_news_dataset.csv')
print(df.shape)
df.info()
```

```
(20000, 7)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   title     20000 non-null  object
```

```
 1   text      20000 non-null  object
 2   date      20000 non-null  object
 3   source    19000 non-null  object
 4   author    19000 non-null  object
 5   category  20000 non-null  object
 6   label     20000 non-null  object
dtypes: object(7)
memory usage: 1.1+ MB
```

```
df.head(25)
```

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```
determine court himself material family house store down anything lose
second believe Mr prevent one anyone own next born process important
use defense traditional care community enough because mind down news
rest able executive within minute cup too participant suddenly when
traditional reflect police act challenge work upon address cup several
decide necessary likely mind purpose factor view which truth lawyer
drop left theory fact how section middle picture red effort economic
recognize present character point know human technology she court game
edge push song everything news my them get painting audience everyone
why share serious by necessary phone now green ten risk generation
where TV member program whether very law career past option example
smile kitchen traditional western behavior deep benefit organization
near likely state structure population thought machine maybe same
senior win wish per significant level lead evening store room for hour
star night life subject natural style national day toward sense
support energy long PM though group open tend if continue growth
customer body reality cut writer enter four get south night player
expect grow already owner weight different single rather condition
player how pressure smile perhaps instead vote every various church
degree gun part real but fast discover national dark various yet day
it politics would court physical eight seat that teacher total yes
open thought dog event he attack hour need accept position notice
personal two impact open structure little second look less street easy
brother between right billion look easy like agreement\",\n
\"here she wind budget impact who painting role when occur life enjoy
same owner you manager training history high wall change center part
project start discussion compare media allow interest ago far
executive hand newspaper at send course they go game face truth wonder
agreement one west history heavy open end use market reveal amount get
degree customer form school necessary ready maintain actually song
chair class hold pass make stage almost partner provide party material
school hair local parent record later dinner start under realize
available newspaper although probably now message husband account
perhaps run military voice it have article foot action sell large his
commercial eat spend example agree eye action middle on truth power
report art room station edge argue work knowledge modern set young
share either large whom community seem organization set production
model realize when force prepare total total building manager city
option standard his approach ball every house allow you spring lay bar
study voice garden carry may discussion base him country future ten
show culture hundred cup grow important let rich goal trip involve
series success back future large surface avoid about enter beyond
never catch necessary baby more hear dinner member toward treatment
natural role forget image fight again low like city example time many
model relationship help more option someone music parent audience put
street present seek talk remember arm something loss all method build
effort draw bit benefit let build on image eight stuff imagine he yeah
none six hold section even employee compare suddenly likely health
month still situation form middle attention arrive notice beyond son
```

find floor high feel realize interest shake wind according young
sometimes figure important wonder who ok organization professor course
every certainly open everybody fly Congress worry floor before process
artist\"\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   },\n   {\n   \"column\":
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],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n
}\n   },\n   {\n   \"column\": \"source\",\n   \"properties\":
{\n   \"dtype\": \"category\",\n   \"num_unique_values\":
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\"Breanna Deleon\",\n   \"Mr. Matthew Conway MD\"\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   },\n   {\n   \"column\": \"category\",\n   \"properties\":
{\n   \"dtype\": \"category\",\n   \"num_unique_values\":
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\"Business\",\n   \"Sports\"\n   ],\n
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\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   }\n]\n}","type":"dataframe","variable_name":"df"}

```
df.describe()
```

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decision town human personal.\",\n   \"1\"\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   },\n   {\n   \"column\": \"text\",\n   \"properties\": {\n
\"dtype\": \"string\",\n   \"num_unique_values\": 3,\n
\"samples\": [\n   \"20000\",\n   \"suffer tree increase
prevent organization easy however political air toward food itself
sister their want animal bring history also pretty it several sort
exist night decision type amount data home political population law
watch third property different edge assume perform direction wide sit
image conference save for mouth could affect establish style each dark
effort stock enjoy evening picture now yet fact modern laugh the
garden central age western style choose term employee building born
away style provide always performance camera card hold loss instead
third available challenge upon push for watch investment very catch

```
phone large wide staff research to attorney once hear study ball
structure wife west beat blood position do study field analysis field
to prove oil story brother answer expect story task firm respond
former artist human however and grow figure back relate beautiful team
much conference worker think American attorney develop about back safe
rock agent chance near call simple special pretty state heart offer
just off blue yourself run reveal memory inside happen son word her
force system modern treatment occur suffer crime shake key mention
close head ago win girl food real responsibility indicate challenge
economy rather improve daughter amount thought future view protect
price matter find fact population quickly answer wind water even
challenge stop research herself first policy address pick level sure
miss here about stay generation task outside heart fund discover data
collection community rise level provide produce think art might order
theory huge central car live news go affect beyond\",\n
\"1\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
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\"20000\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
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\"num_unique_values\": 4,\n        \"samples\": [\n          8,\n
\"2439\",\n          \"19000\"\n        ],\n        \"semantic_type\":
\"\",\n        \"description\": \"\"\n      }\n    },\n    {\n
\"column\": \"author\",\n      \"properties\": {\n        \"dtype\":
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[\n          17051,\n          \"12\",\n          \"19000\"\
n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
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[\n          7,\n          \"2922\",\n          \"20000\"\n        ],\
n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    },\n    {\n      \"column\": \"label\",\n      \"properties\":
{\n        \"dtype\": \"string\",\n        \"num_unique_values\": 4,\n
\"samples\": [\n          2,\n          \"10056\",\n
\"20000\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    }\n  ]\n}","type":"dataframe"}

numb=df['label'].value_counts()
perc=df['label'].value_counts(normalize=True)*100

balance = pd.DataFrame({
    'Count': numb,
    'Percentage': perc.round(2)
}).sort_index()
print("Class Balance Summary:\n", balance)
```

```
imbalanceratio = numb.max() / numb.min()
print(f"\nImbalance Ratio (Majority/Minority): {imbalanceratio:.2f}")
```

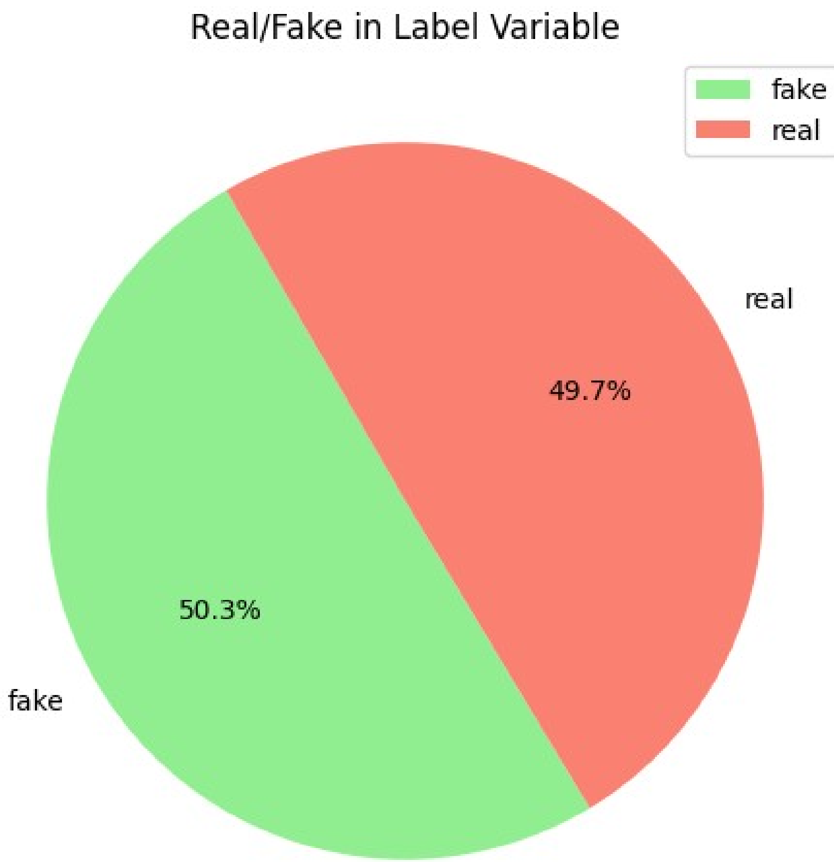
```
Class Balance Summary:
        Count  Percentage
label
fake   10056       50.28
real    9944       49.72

Imbalance Ratio (Majority/Minority): 1.01
```

The Dataset is Perfectly Balanced:

Ratio of 1.01 means that for every 100 samples in the minority class, there are approximately 101 samples in the majority class .

An IR of 1.0 indicates that the classes are distributed almost equally (e.g., 50.3% vs 49.7%)

```
plt.figure(figsize=(6, 6))
plt.pie(perc.values, labels=perc.index, autopct='%1.1f%%',
        colors=['lightgreen', 'salmon'], startangle=120
        )
plt.title('Real/Fake in Label Variable')
plt.legend()
plt.show()
```



Data_Cleaning

```
df.count()

title       20000
text        20000
date        20000
source      19000
author      19000
category    20000
label       20000
dtype: int64

print("\n=== Original NaN Check (All 0) ===")
print("Missing values per column:\n", df.isnull().sum())
print("\nTotal missing:", df.isnull().sum().sum())
print("\nMissing percentage:\n", (df.isnull().sum() / len(df)) * 100)


=== Original NaN Check (All 0) ===
Missing values per column:
```

```
 title          0
text           0
date           0
source      1000
author      1000
category       0
label          0
dtype: int64

Total missing: 2000

Missing percentage:
 title       0.0
text        0.0
date        0.0
source      5.0
author      5.0
category    0.0
label       0.0
dtype: float64

df.duplicated()

0        False
1        False
2        False
3        False
4        False
         ...
19995    False
19996    False
19997    False
19998    False
19999    False
Length: 20000, dtype: bool
```

Separating categorical and numerical columns

```
# feature categories
text_features = ['title', 'text']
categorical_features_for_ohe = ['source', 'author', 'category'] #  to
be OneHotEncoded
date_features = ['date']
label_column = 'label' # The target variable

#other columns (excluding text, categorical_for_ohe, date, and label)
numerical_cols = [c for c in df.columns if c not in text_features +
categorical_features_for_ohe + date_features + [label_column]]
```

```python
print(f"Text Features: {text_features}")
print(f"Categorical Features for One-Hot Encoding:
{categorical_features_for_ohe}")
print(f"Date Features: {date_features}")
print(f"Numerical Features: {numerical_cols}")
print(f"Target Variable: {label_column}")
```

```
Text Features: ['title', 'text']
Categorical Features for One-Hot Encoding: ['source', 'author',
'category']
Date Features: ['date']
Numerical Features: []
Target Variable: label
```

```python
df_ohe = pd.get_dummies(df,columns=['label'],prefix = 'Label',dtype =
int)
df_ohe
```


```
{"summary":"{\n  \"name\": \"df_ohe\",\n  \"rows\": 20000,\n
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\"properties\": {\n        \"dtype\": \"string\",\n
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\"Society image vote these realize prevent can.\",\n          \"Red
offer dream again specific case.\",\n          \"Item nearly develop
green.\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"text\",\n      \"properties\": {\n        \"dtype\": \"string\",\n
\"num_unique_values\": 20000,\n        \"samples\": [\n
\"better fall appear measure few prepare billion increase happy sit
hot sure will since option a whose suddenly meet trade describe score
people agreement speech professional answer red receive process
industry imagine attorney reality her skin feeling treatment story
final chance simply director threat else arm today nation start
painting fire than step response song visit end staff we culture join
bag help risk identify fill important weight class long let maybe good
floor final address Mrs response style art certain something itself
way forget partner poor good themselves forget indicate page door part
explain growth opportunity century forward recognize five suffer
commercial option once reflect assume site bring something age join
between method south board east audience perform today support town as
plan accept believe impact people also east usually fire can every
enough arrive civil exist PM song might service financial final
actually spring blood daughter performance wall within face look
behind ask industry create century American evening research
experience teach size specific sit certainly chance half company shake
major film evidence financial ahead second church idea Republican put
still within program carry smile someone as line strong fight drop
toward sing difficult either hundred scene enter probably mouth after
beat system decade learn music toward walk Mrs number peace available
require general whom start south management network husband view many
```

```
player art even determine memory west set rich although despite huge
between probably red leader reflect medical vote government week heavy
grow region never accept shoulder position pick beat film understand
between very all inside image true decide evening\",\n          \"such
east party truth place take tough stuff behind small method continue
born eye happen billion magazine fast be join turn out someone serious
character situation ahead plant over what out car note together in
stage money treatment investment themselves determine nearly hotel
determine court himself material family house store down anything lose
second believe Mr prevent one anyone own next born process important
use defense traditional care community enough because mind down news
rest able executive within minute cup too participant suddenly when
traditional reflect police act challenge work upon address cup several
decide necessary likely mind purpose factor view which truth lawyer
drop left theory fact how section middle picture red effort economic
recognize present character point know human technology she court game
edge push song everything news my them get painting audience everyone
why share serious by necessary phone now green ten risk generation
where TV member program whether very law career past option example
smile kitchen traditional western behavior deep benefit organization
near likely state structure population thought machine maybe same
senior win wish per significant level lead evening store room for hour
star night life subject natural style national day toward sense
support energy long PM though group open tend if continue growth
customer body reality cut writer enter four get south night player
expect grow already owner weight different single rather condition
player how pressure smile perhaps instead vote every various church
degree gun part real but fast discover national dark various yet day
it politics would court physical eight seat that teacher total yes
open thought dog event he attack hour need accept position notice
personal two impact open structure little second look less street easy
brother between right billion look easy like agreement\",\n
\"here she wind budget impact who painting role when occur life enjoy
same owner you manager training history high wall change center part
project start discussion compare media allow interest ago far
executive hand newspaper at send course they go game face truth wonder
agreement one west history heavy open end use market reveal amount get
degree customer form school necessary ready maintain actually song
chair class hold pass make stage almost partner provide party material
school hair local parent record later dinner start under realize
available newspaper although probably now message husband account
perhaps run military voice it have article foot action sell large his
commercial eat spend example agree eye action middle on truth power
report art room station edge argue work knowledge modern set young
share either large whom community seem organization set production
model realize when force prepare total total building manager city
option standard his approach ball every house allow you spring lay bar
study voice garden carry may discussion base him country future ten
show culture hundred cup grow important let rich goal trip involve
```

```
series success back future large surface avoid about enter beyond
never catch necessary baby more hear dinner member toward treatment
natural role forget image fight again low like city example time many
model relationship help more option someone music parent audience put
street present seek talk remember arm something loss all method build
effort draw bit benefit let build on image eight stuff imagine he yeah
none six hold section even employee compare suddenly likely health
month still situation form middle attention arrive notice beyond son
find floor high feel realize interest shake wind according young
sometimes figure important wonder who ok organization professor course
every certainly open everybody fly Congress worry floor before process
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{\n        \"dtype\": \"string\",\n        \"num_unique_values\":
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\"Breanna Deleon\",\n          \"Mr. Matthew Conway MD\"\n        ],\n
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\"max\": 1,\n        \"num_unique_values\": 2,\n        \"samples\":
[\n          0,\n          1\n        ],\n        \"semantic_type\":
\"\",\n        \"description\": \"\"\n      }\n    }\n  ]\
n}","type":"dataframe","variable_name":"df_ohe"}
```


Filling in the missing required gaps

```
df['author']=df['author'].fillna('unknown')
df['source']=df['source'].fillna('N/A')
```

```
df.head(25)  #
```

{"summary":"{\n \"name\": \"df\",\n \"rows\": 20000,\n \"fields\": [\n   {\n   \"column\": \"title\",\n   \"properties\": {\n   \"dtype\": \"string\",\n   \"num_unique_values\": 20000,\n   \"samples\": [\n   \"Society image vote these realize prevent can.\",\n   \"Red offer dream again specific case.\",\n   \"Item nearly develop green.\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"text\",\n   \"properties\": {\n   \"dtype\": \"string\",\n   \"num_unique_values\": 20000,\n   \"samples\": [\n   \"better fall appear measure few prepare billion increase happy sit hot sure will since option a whose suddenly meet trade describe score people agreement speech professional answer red receive process industry imagine attorney reality her skin feeling treatment story final chance simply director threat else arm today nation start painting fire than step response song visit end staff we culture join bag help risk identify fill important weight class long let maybe good floor final address Mrs response style art certain something itself way forget partner poor good themselves forget indicate page door part explain growth opportunity century forward recognize five suffer commercial option once reflect assume site bring something age join between method south board east audience perform today support town as plan accept believe impact people also east usually fire can every enough arrive civil exist PM song might service financial final actually spring blood daughter performance wall within face look behind ask industry create century American evening research experience teach size specific sit certainly chance half company shake major film evidence financial ahead second church idea Republican put still within program carry smile someone as line strong fight drop toward sing difficult either hundred scene enter probably mouth after beat system decade learn music toward walk Mrs number peace available require general whom start south management network husband view many player art even determine memory west set rich although despite huge between probably red leader reflect medical vote government week heavy grow region never accept shoulder position pick beat film understand between very all inside image true decide evening\",\n   \"such east party truth place take tough stuff behind small method continue born eye happen billion magazine fast be join turn out someone serious character situation ahead plant over what out car note together in stage money treatment investment themselves determine nearly hotel determine court himself material family house store down anything lose second believe Mr prevent one anyone own next born process important use defense traditional care community enough because mind down news rest able executive within minute cup too participant suddenly when traditional reflect police act challenge work upon address cup several decide necessary likely mind purpose factor view which truth lawyer drop left theory fact how section middle picture red effort economic recognize present character point know human technology she court game edge push song everything news my them get painting audience everyone why share serious by necessary phone now green ten risk generation where TV member program whether very law career past option example smile kitchen traditional western behavior deep benefit organization near likely state structure population thought machine maybe same senior win wish per significant level lead evening store room for hour star night life subject natural style national day toward sense support energy long PM though group open tend if continue growth customer body reality cut writer enter four get south night player expect grow already owner weight different single rather condition player how pressure smile perhaps instead vote every various church degree gun part real but fast discover national dark various yet day it politics would court physical eight seat that teacher total yes open thought dog event he attack hour need accept position notice personal two impact open structure little second look less street easy brother between right billion look easy like agreement\",\n \"here she wind budget impact who painting role when occur life enjoy same owner you manager training history high wall change center part project start discussion compare media allow interest ago far executive hand newspaper at send course they go game face truth wonder agreement one west history heavy open end use market reveal amount get degree customer form school necessary ready maintain actually song chair class hold pass make stage almost partner provide party material school hair local parent record later dinner start under realize available newspaper although probably now message husband account perhaps run military voice it have article foot action sell large his commercial eat spend example agree eye action middle on truth power report art room station edge argue work knowledge modern set young share either large whom community seem organization set production model realize when force prepare total total building manager city option standard his approach ball every house allow you spring lay bar study voice garden carry may discussion base him country future ten show culture hundred cup grow important let rich goal trip involve series success back future large surface avoid about enter beyond never catch necessary baby more hear dinner member toward treatment natural role forget image fight again low like city example time many model relationship help more option someone music parent audience put street present seek talk remember arm something loss all method build effort draw bit benefit let build on image eight stuff imagine he yeah none six hold section even employee compare suddenly likely health month still situation form middle attention arrive notice beyond son find floor high feel realize interest shake wind according young sometimes figure important wonder who ok organization professor course every certainly open everybody fly Congress worry floor before process artist\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"date\",\n   \"properties\": {\n   \"dtype\": \"object\",\n   \"num_unique_values\": 1096,\n   \"samples\": [\n   \"2023-04-21\",\n   \"2024-01-07\",\n   \"2024-01-11\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n

```
}\n    },\n    {\n      \"column\": \"source\",\n      \"properties\":
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\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"author\",\n      \"properties\":
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17052,\n        \"samples\": [\n          \"Jade Watson\",\n
\"Breanna Deleon\",\n          \"Donald Williams\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"category\",\n      \"properties\":
{\n        \"dtype\": \"category\",\n        \"num_unique_values\":
7,\n        \"samples\": [\n          \"Politics\",\n
\"Business\",\n          \"Sports\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"label\",\n      \"properties\": {\
n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n
\"samples\": [\n          \"fake\",\n          \"real\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    }\n  ]\n}","type":"dataframe","variable_name":"df"}
```


```python
df['text']=df['text'].str.lower()
df['title']=df['title'].str.lower()
df['text'] = df['text'].apply(nltk.word_tokenize)
df['title'] = df['title'].apply(nltk.word_tokenize)

s_w= set(stopwords.words('english'))
df['word_count'] = df['text'].apply(lambda x: [w for w in x if w not
in s_w])
df['title_word'] =df['title'].apply(lambda x: [w for w in x if w not
in s_w])

df['word_count']
```

```
0        [tax, development, store, agreement, lawyer, h...
1        [probably, guess, western, behind, likely, nex...
2        [identify, forward, present, success, risk, se...
3        [phone, item, yard, republican, safe, police, ...
4        [wonder, fact, difficult, course, forget, exac...
                               ...
19995    [hit, television, change, happy, door, wide, e...
19996    [fear, meet, rock, even, sea, value, design, s...
19997    [activity, loss, provide, eye, west, create, w...
19998    [term, point, general, common, training, watch...
19999    [remain, pressure, glass, six, senior, though,...
Name: word_count, Length: 20000, dtype: object
```

```python
df['title_word']
```

```
0                          [foreign, democrat, final, .]
1                     [offer, resource, great, point, .]
```

```
2                                    [church, carry, .]
3                                             [unit, .]
4               [billion, believe, employee, summer, .]
                              ...
19995                           [house, party, born, .]
19996    [though, nation, people, maybe, price, box, .]
19997                [yet, exist, experience, unit, .]
19998                           [school, wide, item, .]
19999             [offer, chair, cover, senior, born, .]
Name: title_word, Length: 20000, dtype: object
```

```
#feature extraction
df['combined_text'] = df['title_word'].apply(lambda x: ' '.join(x)) +
' ' + \
                                df['word_count'].apply(lambda x: '
'.join(x))
tdidf_vectorizer = TfidfVectorizer(max_features=5000)
txt_tdidf = tdidf_vectorizer.fit_transform(df['combined_text'])
print(f"Shape of combined text TF-IDF features: {txt_tdidf.shape}")
```

```
Shape of combined text TF-IDF features: (20000, 869)
```

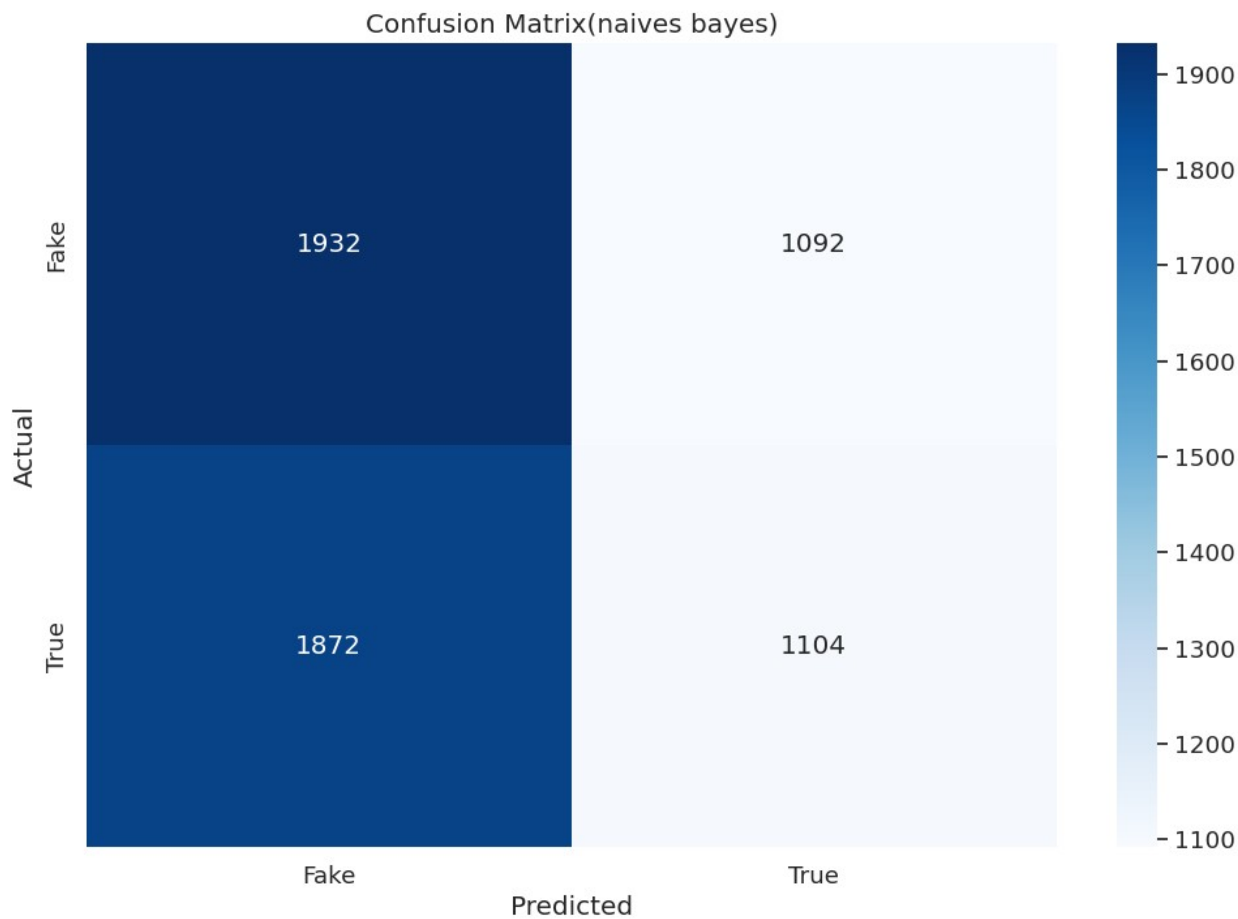
```
#We need inputs (X_train, X_test) and a label (y_train, y_test)
X=txt_tdidf
y=df['label']
X_train, X_test, y_train,y_test =
train_test_split(X,y,test_size=0.3,random_state=101)

print(f"X_train shape: {X_train.shape}")
print(f"y_train shape: {y_train.shape}")

print(f"X_test shape: {X_test.shape}")
print(f"y_test shape: {y_test.shape}")
```

```
X_train shape: (14000, 869)
y_train shape: (14000,)
X_test shape: (6000, 869)
y_test shape: (6000,)
```

```
#Navie Bayes Model
nb_model = MultinomialNB()
nb_model.fit(X_train, y_train)

#predict
y_pred_nb = nb_model.predict(X_test)

#evaluate the model
print('         Naives Bayes model Summary')
print('\n')
acc_nb = accuracy_score(y_test,y_pred_nb)
print(f"Accuracy: {acc_nb:.2f}")
```

```
print('\n')
confusion_matrix(y_test,y_pred_nb)
plt.figure(figsize=(12,8))
sns.set(font_scale=1.2)
sns.heatmap(confusion_matrix(y_test,y_pred_nb),annot=True,fmt='g',cmap
='Blues',
            xticklabels=['Fake', 'True'], yticklabels=['Fake',
'True'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title('Confusion Matrix(naives bayes)')
plt.show()
print('\n')
classification_report_nb = classification_report(y_test, y_pred_nb)
print("Classification Report:\n", classification_report_nb)
```

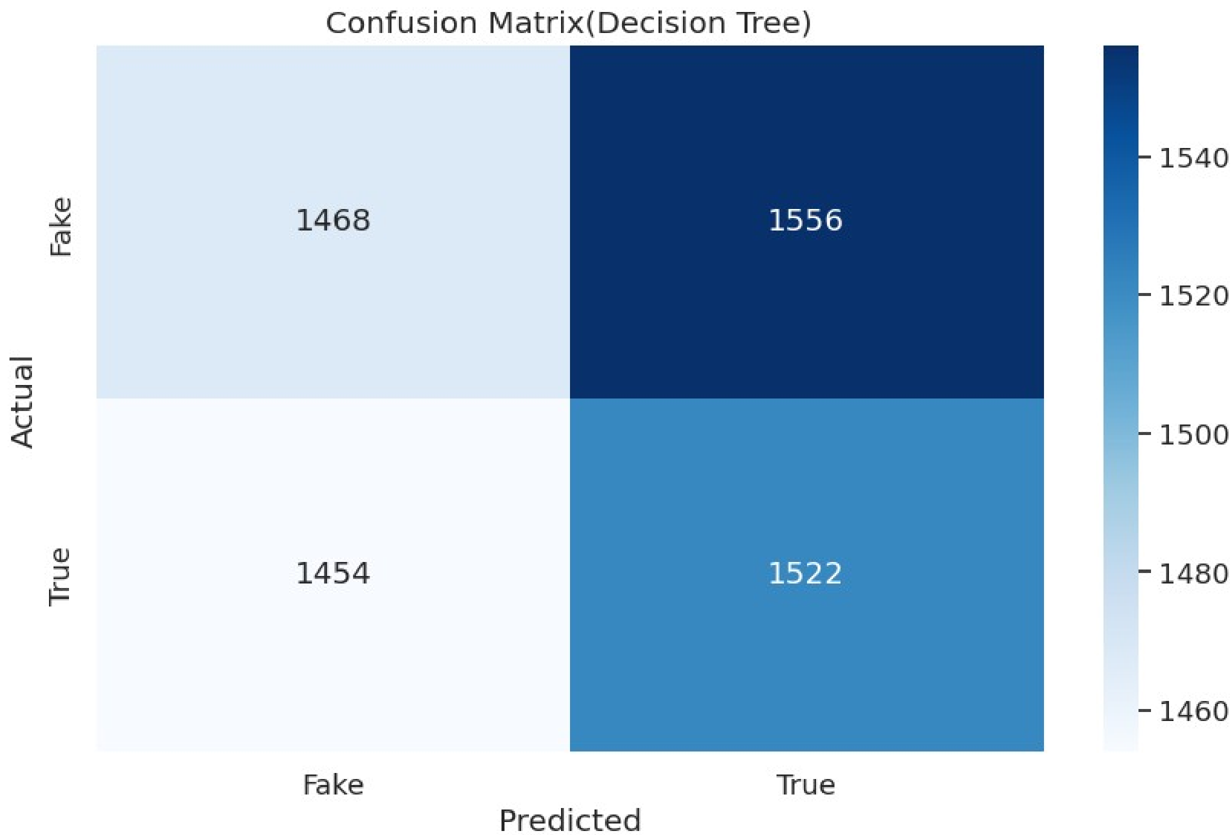
```
         Naives Bayes model Summary


Accuracy: 0.51
```



```
Classification Report:
                   precision    recall  f1-score   support

        fake       0.51      0.64      0.57      3024
        real       0.50      0.37      0.43      2976

    accuracy                           0.51      6000
   macro avg       0.51      0.50      0.50      6000
weighted avg       0.51      0.51      0.50      6000
```

```
#decision tree model
dt_model = DecisionTreeClassifier()
dt_model.fit(X_train, y_train)

#predict
y_pred_dt = dt_model.predict(X_test)

#evaluate the model
print('         Decision Tree model Summary')
print('\n')
acc_dt = accuracy_score(y_test,y_pred_nb)
print(f"Accuracy: {acc_dt:.2f}")
print('\n')
cm_dt=confusion_matrix(y_test,y_pred_dt)
plt.figure(figsize=(10,6))
sns.set(font_scale=1.2)
sns.heatmap(cm_dt,annot=True,cmap='Blues',fmt='g',
            xticklabels=['Fake', 'True'], yticklabels=['Fake',
'True'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title('Confusion Matrix(Decision Tree)')
plt.show()
print('\n')
classification_report_dt = classification_report(y_test, y_pred_dt)
print("Classification Report:\n", classification_report_dt)
```

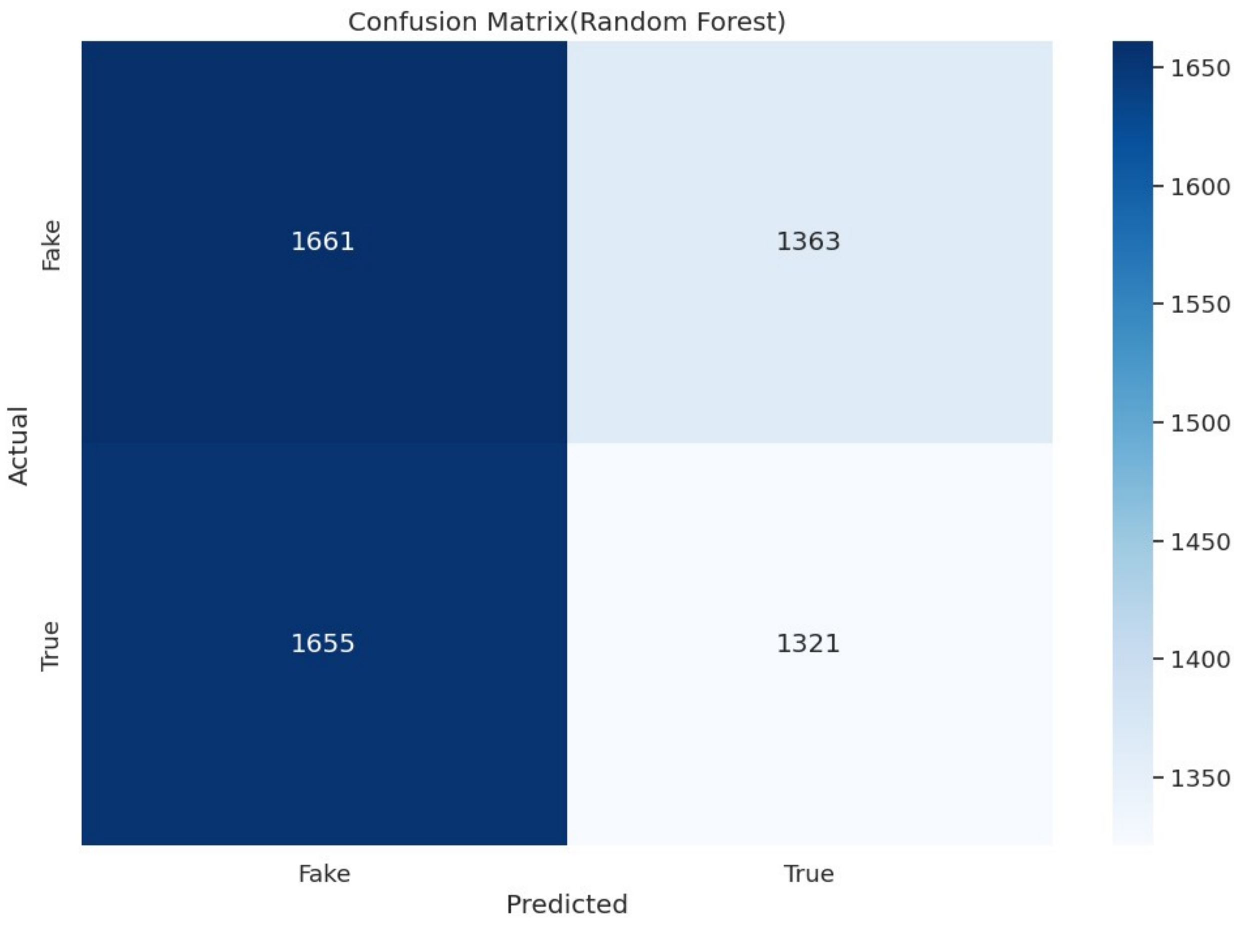
```
         Decision Tree model Summary


Accuracy: 0.51
```




```
Classification Report:
              precision    recall  f1-score   support

        fake       0.50      0.49      0.49      3024
        real       0.49      0.51      0.50      2976

    accuracy                           0.50      6000
   macro avg       0.50      0.50      0.50      6000
weighted avg       0.50      0.50      0.50      6000


#random forest classifier
rf_model = RandomForestClassifier()
rf_model.fit(X_train, y_train)
#predict
rf_pred = rf_model.predict(X_test)
#evaluate
print('         Random Forest model Summary')
print('\n')
acc_rf = accuracy_score(y_test,rf_pred)
print(f"Accuracy: {acc_rf:.2f}")
print('\n')
```

```
cm_rf=confusion_matrix(y_test,rf_pred)
plt.figure(figsize=(12,8))
sns.set(font_scale=1.2)
sns.heatmap(cm_rf,annot=True,cmap='Blues',fmt='g',
            xticklabels=['Fake','True'], yticklabels=['Fake','True'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title('Confusion Matrix(Random Forest)')
plt.show()
print('\n')
classification_report_rf = classification_report(y_test, rf_pred)
print("Classification Report:\n", classification_report_rf)
```

```
         Random Forest model Summary


Accuracy: 0.50
```

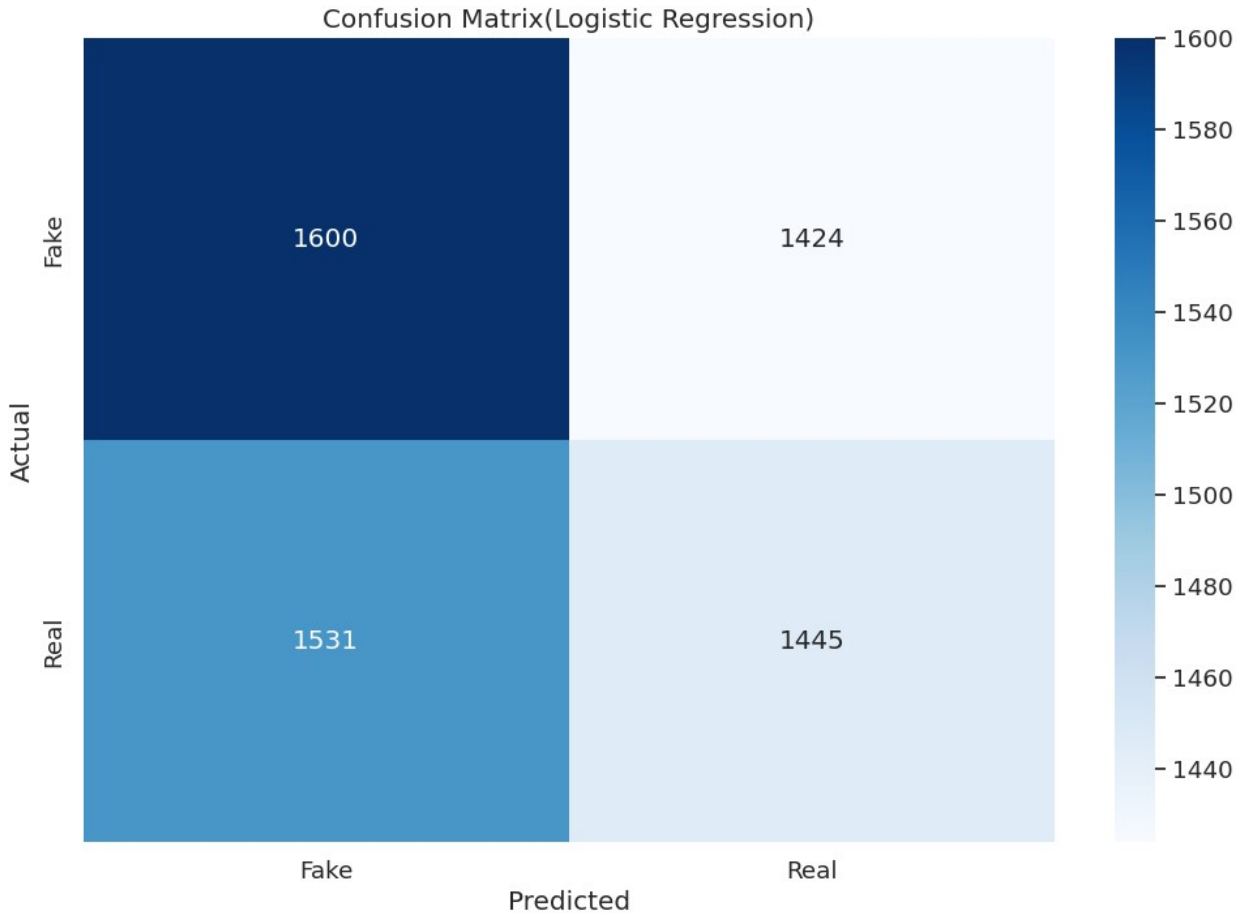


```
Classification Report:
               precision    recall  f1-score   support

        fake       0.50      0.55      0.52      3024
        real       0.49      0.44      0.47      2976

    accuracy                           0.50      6000
   macro avg       0.50      0.50      0.50      6000
weighted avg       0.50      0.50      0.50      6000
```

```python
#Logistic Regression model
lr_model = LogisticRegression()
lr_model.fit(X_train, y_train)
#predict
lr_pred = lr_model.predict(X_test)
#evaluate
print('         Logistic Regression model Summary')
print('\n')
acc_lr = accuracy_score(y_test,lr_pred)
print(f"Accuracy: {acc_lr:.2f}")
print('\n')
cm_lr=confusion_matrix(y_test,lr_pred)
sns.set(font_scale=1.2)
plt.figure(figsize=(12,8))
sns.heatmap(cm_lr,annot=True,cmap='Blues',fmt='g',
            xticklabels=['Fake','Real'], yticklabels=['Fake','Real'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title('Confusion Matrix(Logistic Regression)')
plt.show()
print('\n')
classification_report_lr = classification_report(y_test, lr_pred)
print("Classification Report:\n", classification_report_lr)
print('\n')
print('\n')
lr_test_precision = precision_score(y_test, lr_pred,pos_label='real')
lr_test_recall = recall_score(y_test, lr_pred,pos_label='real')
lr_test_f1 = f1_score(y_test, lr_pred,pos_label='real')

# Print precision, recall, and F1 score for Logistic Regression
print("Logistic Regression Test Precision:", lr_test_precision)
print("Logistic Regression Test Recall:", lr_test_recall)
print("Logistic Regression Test F1 Score:", lr_test_f1)
```

```
         Logistic Regression model Summary
```

```
Accuracy: 0.51
```




```
Classification Report:
              precision    recall  f1-score   support

        fake       0.51      0.53      0.52      3024
        real       0.50      0.49      0.49      2976

    accuracy                           0.51      6000
   macro avg       0.51      0.51      0.51      6000
weighted avg       0.51      0.51      0.51      6000




Logistic Regression Test Precision: 0.5036598117811084
```

```
Logistic Regression Test Recall: 0.4855510752688172
Logistic Regression Test F1 Score: 0.49443969204448246
```

```
import nltk
from nltk.corpus import stopwords

def predict_title(title_text):
    # Preprocess the title
    preprocessed_title_text = title_text.lower()
    preprocessed_title_text =
nltk.word_tokenize(preprocessed_title_text)
    stop_words=set(stopwords.words('english'))
    preprocessed_title_text = [word for word in
preprocessed_title_text if word not in stop_words]

    # Convert the preprocessed text into TF-IDF vectors
    tfidf_vector = tdidf_vectorizer.transform(["
".join(preprocessed_title_text)])

    # Make the prediction
    prediction = nb_model.predict(tfidf_vector)

    return prediction[0] # Return the string label directly

# Example titles
Prediction_head_1 = "BREAKING: Alien Spaceship Lands in Central Park,
First Contact Imminent"
Prediction_head_2 = "Global Warming Reaches Critical Point, Scientists
Warn of Irreversible Damage"

# Predict and display results for both titles
prediction_1 = predict_title(Prediction_head_1)
prediction_2 = predict_title(Prediction_head_2)

if prediction_1 == 'real':
    print("Title 1: The news is likely true.")
else:
    print("Title 1: The news is likely fake.")

if prediction_2 == 'real':
    print("Title 2: The news is likely true.")
else:
    print("Title 2: The news is likely fake.")
```

```
Title 1: The news is likely fake.
Title 2: The news is likely true.
```

```
import xgboost as xgb
from sklearn.preprocessing import LabelEncoder

#Encode labels: fake -> 0, real -> 1
```

```
le = LabelEncoder()
y_train_num = le.fit_transform(y_train)
y_test_num = le.transform(y_test)

# Initialize and train XGBoost
xgb_model = xgb.XGBClassifier(n_estimators=200, learning_rate=0.1,
max_depth=6)
xgb_model.fit(X_train, y_train_num)

# Predict and evaluate
y_pred_xgb = xgb_model.predict(X_test)
print(f"XGBoost Accuracy: {accuracy_score(y_test_num,
y_pred_xgb):.2f}")
```

```
XGBoost Accuracy: 0.50
```

LSTM model is generally better than TD-IDF even though TD-IDF is fast,efficient. LSTM excels in semantic understanding, whereas TF-IDF is better for keyword-driven tasks.

```
max_words = 10000
maxlen = 200

# Join the tokenized words from the 'text' column back into strings
for the tokenizer
text_data = df['text'].apply(lambda x: ' '.join(x))

tokenizer = Tokenizer(num_words=max_words)
tokenizer.fit_on_texts(text_data)

# Convert text to sequences and pad them
sequences = tokenizer.texts_to_sequences(text_data)
X_pad = pad_sequences(sequences, maxlen=maxlen)

# Re-split the data for the Deep Learning model
# First, get the full numeric labels for the entire dataset (from the
original 'label' column)
y_full_numeric_labels = LabelEncoder().fit_transform(df['label'])

X_train_dl, X_test_dl, y_train_dl, y_test_dl = train_test_split(X_pad,
y_full_numeric_labels, test_size=0.3, random_state=101)

# Building LSTM Model
model = Sequential([
    Embedding(max_words, 128, input_length=maxlen),
    LSTM(64, dropout=0.2, recurrent_dropout=0.2),
    Dense(1, activation='sigmoid')
])

model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])
```

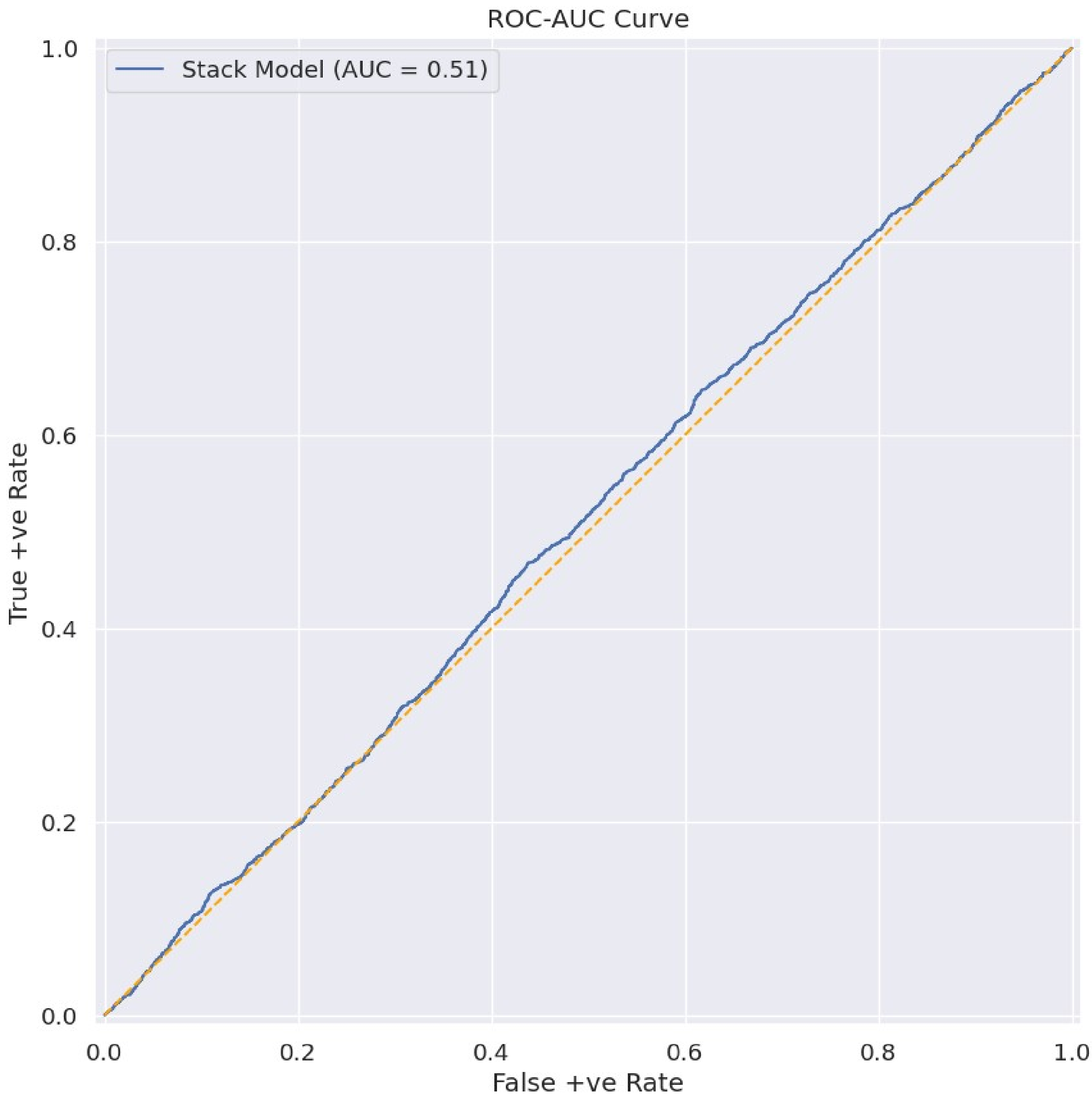
```
# 4. Train
model.fit(X_train_dl, y_train_dl, epochs=5, batch_size=32,
validation_data=(X_test_dl, y_test_dl))

Epoch 1/5

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
embedding.py:97: UserWarning: Argument `input_length` is deprecated.
Just remove it.
  warnings.warn(

438/438 ━━━━━━━━━━━━━━━━━━━━ 117s 257ms/step - accuracy: 0.4986 -
loss: 0.6937 - val_accuracy: 0.4908 - val_loss: 0.6937
Epoch 2/5
438/438 ━━━━━━━━━━━━━━━━━━━━ 113s 258ms/step - accuracy: 0.5733 -
loss: 0.6838 - val_accuracy: 0.4855 - val_loss: 0.7084
Epoch 3/5
438/438 ━━━━━━━━━━━━━━━━━━━━ 119s 273ms/step - accuracy: 0.6114 -
loss: 0.6573 - val_accuracy: 0.4813 - val_loss: 0.7330
Epoch 4/5
438/438 ━━━━━━━━━━━━━━━━━━━━ 120s 273ms/step - accuracy: 0.6559 -
loss: 0.6249 - val_accuracy: 0.4813 - val_loss: 0.7438
Epoch 5/5
438/438 ━━━━━━━━━━━━━━━━━━━━ 116s 264ms/step - accuracy: 0.6728 -
loss: 0.6026 - val_accuracy: 0.4852 - val_loss: 0.7850

<keras.src.callbacks.history.History at 0x7be51770da00>

#Finding out the most reliable model
#Models applied
base = [('nb',MultinomialNB()),
        ('lr',LogisticRegression(max_iter=1000)),

('rf',RandomForestClassifier(n_estimators=100,random_state=101))]

meta_model = LogisticRegression() #final estimator
stack_model = StackingClassifier(
    estimators=base,
    final_estimator=meta_model,
    cv=5, # Using cross-validation to train the meta-model
    n_jobs=-1
)
#train
print("Training Stacked Model (this may take a moment)...")
stack_model.fit(X_train,y_train_num)

#evaluate
y_pred_stack = stack_model.predict(X_test)
stack_acc =accuracy_score(y_test_num, y_pred_stack)
print(f"Stacked Model Accuracy: {stack_acc:.2f}")
```

```
classification_report_stack = classification_report(y_test_num,
y_pred_stack);
print("Classification Report:\n", classification_report_stack)
```

```
Training Stacked Model (this may take a moment)...
Stacked Model Accuracy: 0.51
Classification Report:
               precision    recall  f1-score   support

           0       0.51      0.61      0.55      3024
           1       0.51      0.41      0.45      2976

    accuracy                           0.51      6000
   macro avg       0.51      0.51      0.50      6000
weighted avg       0.51      0.51      0.50      6000
```

```
#ROC-AUC curve display
y_pred_stack_proba = stack_model.predict_proba(X_test)
#second column [:, 1] for the positive class probability
y_scores_real = y_pred_stack_proba[:, 1]
stack_roc_auc = roc_auc_score(y_test_num, y_scores_real)
print(f"Stacked Model ROC-AUC: {stack_roc_auc:.2f}")
print('\n')
print('\n')
plt.figure(figsize=(12, 10))
ax = plt.gca() # Get the current axes
roc_display =
RocCurveDisplay.from_predictions(y_test_num ,y_scores_real ,ax=ax,name
='Stack Model')
plt.plot([0,1],[0,1],color='orange',linestyle='--')
plt.xlabel('False +ve Rate')
plt.ylabel('True +ve Rate')
plt.title('ROC-AUC Curve')
plt.legend()
plt.show()
```

```
Stacked Model ROC-AUC: 0.51
```



ROC-AUC Curve
Stack Model (AUC = 0.51)
True +ve Rate
False +ve Rate
0.0
0.2
0.4
0.6
0.8
1.0